

Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

Is that a virus?



Learning Outcome:

Uses the four basic functions in persistent storage to create the system. Talks about the four basic functions for persistent storage.

Ask your partner:

- Why do you think safety is important at the moment of programming?
- Why do you think a person would hack a database?

Can Do Statements:

- I can tell why using CRUD is better than using other statements.
- I can explain how to create a new record using CRUD.
- I can talk about how to update a record.





Let's Explore!

Sharon is helping John out with a problem. What is it?



Instructions: Read the text below.

John: Hi, Sharon. Do you know what we are doing tomorrow in Mr. Ferguson class?

Sharon: Hmm, let me see, taking into account last class we checked what CRUD was and the creation of databases using MySQL and its relation with CRUD statements. I think tomorrow we are having one of those pop quizzes he likes doing after every single topic.

John: Wow, I missed a lot; it seems I'd better catch up on my studies.

Sharon: If you want, I can help you out with that.

John: Sure, what's all that MySQL thing about?

Sharon: Well, according to Mr. Ferguson, you might want to use it to create the databases with SQL language, as all databases work with it.

John: Interesting, but do you know how to create a database?

Sharon: Sure. It is very simple to process. In fact, if you know what queries to type in, everything will go really smooth; for example, when you are creating a new record, you have to use the INSERT statement to create such data in the database, as well as when you are updating an existing one, you have to use the UPDATE statement based on a the primary key you've previously created.

John: God, that's too much information to process.

Sharon: Chill out, that's nothing you can't handle.

John: Now, why are we using CRUD instead of any other statement?

Sharon: If I'm not wrong, it is because you can use it with something called stored procedures, which is something that will protect the database from SQL injection attacks that can delete the whole database and jeopardize the performance.

John: Well, that's something I ought to study before the exam.



**Let's Practice!**

Why do you choose CRUD?

Instructions: Write the missing words to complete the sentences.

1. You'd better use CRUD instead of any other statement because you can use it with _s_____d _____s_.
2. Sharon might want to protect the database from _____L _____ct_____ attacks by using CRUD.
3. You had better protect the database instead of being in difficult situations trying to _____ev___ the information.
4. He ought to guarantee the security of the _____or_____e_ with a good programmer.
5. The company might want to invest in quality systems to prevent __ac____
____tt_____.

**Let's Practice!**

Some instructions are necessary!

Instructions: Select the correct answer to complete the sentences.

1. I'm _____to you how to create a database, but first, you have to turn on your server.
Hint: This situation is happening right now.
a. explain
b. explaining
c. will explain
d. going to explain
2. She's _____a new record next Saturday.
Hint: This situation is arranged previously.
a. creating
b. create
c. will create
d. going to create



**Let's Practice!**

3. He's _____ the parameters on the database columns to process the information.
Hint: This situation is happening right now.
a. identify
b. will identify
c. identifying
d. going to identify
4. I _____ the records but I really need that you explain me some things before doing it.
Hint: Think about the next week.
a. creating
b. create
c. will create
d. going to create
5. The new employer is _____ the staff's personal information database in the system.
Hint: This is something certain to happen.
a. creating
b. create
c. will create
d. going to create

**Extension Activity**

Why do you choose CRUD?

Instructions: Complete the following steps. Fill in the Self-Assessment.

- a) Read about the importance of using CRUD instead of other statements.
- b) Make a brochure explaining the importance of CRUD over other programs. Describe its functions and key points.
- c) Explain the brochure to your classmates highlighting the most important aspects about CRUD.

Self-Assessment**Yes** **No**

1. I read about CRUD. ☐ ☐
2. I recognize the importance of CRUD over other statements. ☐ ☐
3. I make a brochure about CRUD. ☐ ☐
4. I explain my brochure to my class. ☐ ☐



Target Words

- **query**(noun): instructions to elicit information from a database
e.g. If you know what queries to type in, everything will go really smooth.
- **primary key** (noun): unique identification for a record in a database
e.g. You have to use the UPDATE statement based on the primary key you've previously created.
- **chill out**(phrasal verb): to relax
e.g. Chill out, that's nothing you can't handle.
- **pop quiz**(noun): an unexpected quiz
e.g. I think tomorrow we are having one of those pop quizzes.
- **catch up on**(phrasal verb): to get or give the latest news about something to someone
e.g. It seems I'd better catch up on my studies.
- **MySQL** (noun): open source relational database management system
e.g. What's all that MySQL thing about?



Learning Points

Present Continuous for Future

We can express ideas with **present continuous** to talk about the future.

Present continuous is commonly used to talk about situations that are happening at the moment of talking (ongoing actions), but you can also use it to express ideas in a fixed or arranged future.

When a future idea is fixed or people expect it to happen since it was previously arranged, such idea can be expressed using **present continuous**.

If you want to make a sentence expressing future with present continuous, simply use the structure for such time with a time expression showing future arrangement. "**Subject + to be + verb + ing + complement.**"

Example:

We are coding all those apps tomorrow morning.

Examples

- *Do you know what we are doing tomorrow in Mr. Ferguson class?*
- *I think tomorrow we are having one of those pop quizzes he likes doing after every single topic.*





Learning Points

What is an injection attack?

A **SQL injection attack** is an attack done by hackers to delete, steal or retrieve information from databases.

SQL injection attacks usually take place when programmers build the SQL statements dynamically by joining or concatenating the strings instead of using stored procedures.

When you use a poor design, it can allow a hacker to type some basic SQL codes into a text box in a user log in or a search box of a webpage and retrieve all the users or data information.

In order to prevent such SQL injection attacks, the programmer can use stored procedures where the prefixes differ from other prefixes used in the other user procedures.

